

INTRODUCTION TO SYNTAX I

Session 2: Syntactic structures have requirements to satisfy

Timea Szarvas

University of Potsdam

`timea.szarvas@uni-potsdam.de`

July 28th 2026

OVA 2026

TODAY'S PLAN

- ◉ Refining our notion of hierarchy
- ◉ Smallest units: features
- ◉ Constituents have restrictions
- ◉ Constituents have requirements
- ◉ (Tomorrow's class: how to satisfy these requirements)

A CENTRAL RELATION: C-COMMAND

I What are syntactic processes sensitive to?

- ◉ We already know that syntactic processes target constituents, not random strings.
- ◉ They are also sensitive to the **hierarchical relation** between constituents.

A CENTRAL RELATION: C-COMMAND

What are syntactic processes sensitive to?

- We already know that syntactic processes target constituents, not random strings.
- They are also sensitive to the **hierarchical relation** between constituents.

A very important one of these hierarchical relation is c-command (c for constituent):

⇒ Node α **c-commands** node β and everything β dominates if α and β are sisters.

A CENTRAL RELATION: C-COMMAND

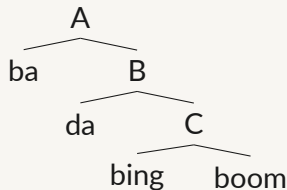
What are syntactic processes sensitive to?

- We already know that syntactic processes target constituents, not random strings.
- They are also sensitive to the **hierarchical relation** between constituents.

A very important one of these hierarchical relation is c-command (c for constituent):

⇒ Node α **c-commands** node β and everything β dominates if α and β are sisters.

- Can you name the c-command relations in this tree?
- (C-command will become really important later on!)



I Quick recap

- ⊙ We established that syntax cares about **constituents** (and c-command).
 - ⊙ We have not yet modeled **why syntactic processes are triggered** in the first place.
- ⇒ Syntactic processes do not just happen randomly in this theory of syntax!

I Quick recap

- ◉ We established that syntax cares about **constituents** (and c-command).
 - ◉ We have not yet modeled **why syntactic processes are triggered** in the first place.
- ⇒ Syntactic processes do not just happen randomly in this theory of syntax!

I Recall again the Y-model from yesterday:

- ◉ The mental lexicon contains all **featural and lexical** elements of a language L.
- ◉ The numeration contains certain **featural and lexical** elements selected from the mental lexicon to build the structure at hand.

I Quick recap

- ◉ We established that syntax cares about **constituents** (and c-command).
 - ◉ We have not yet modeled **why syntactic processes are triggered** in the first place.
- ⇒ Syntactic processes do not just happen randomly in this theory of syntax!

I Recall again the Y-model from yesterday:

- ◉ The mental lexicon contains all **featural and lexical** elements of a language L.
 - ◉ The numeration contains certain **featural and lexical** elements selected from the mental lexicon to build the structure at hand.
- ⇒ Today, we are going to take a closer look at these featural elements.

■ Subject-verb agreement in English

- ⊙ Agreement can be understood as co-variation between two syntactic nodes:

- (1) a. The cat chase**s** the mouse.
b. The cat**s** chase the mouse.

- (2) a. *The cat chase the mouse.
b. *The cat**s** chase**s** the mouse.

■ Subject-verb agreement in English

- ⊙ Agreement can be understood as co-variation between two syntactic nodes:

- (1) a. The cat chase**s** the mouse.
b. The cat**s** chase the mouse.

- (2) a. *The cat chase the mouse.
b. *The cat**s** chase**s** the mouse.

- ⊙ Based on (2), we could hypothesize that agreement makes reference to the *outer appearance* (morphological form) of words.

⇒ Only one -s is allowed when a third person subject agrees with the verb.

■ **Hypothesis 1: If the subject has an -s, the verb does not and vice versa.**

Counterevidence to Hypothesis 1:

- | | | | | | |
|-----|----|-------------------------|-----|----|------------------------------------|
| (3) | a. | The ox eats the hay. | (4) | a. | The child takes piano lessons. |
| | b. | The oxen eat the hay. | | b. | The children take piano lessons. |
| | c. | *The oxen eats the hay. | | c. | *The children takes piano lessons. |

■ Counterevidence to Hypothesis 1:

- | | | | |
|--------|-------------------------|--------|------------------------------------|
| (3) a. | The ox eats the hay. | (4) a. | The child takes piano lessons. |
| b. | The oxen eat the hay. | b. | The children take piano lessons. |
| c. | *The oxen eats the hay. | c. | *The children takes piano lessons. |

■ Hypothesis 2: Agreement makes reference to abstract morphosyntactic features.

■ Counterevidence to Hypothesis 1:

- | | |
|-----------------------------|---------------------------------------|
| (3) a. The ox eats the hay. | (4) a. The child takes piano lessons. |
| b. The oxen eat the hay. | b. The children take piano lessons. |
| c. *The oxen eats the hay. | c. *The children takes piano lessons. |

■ Hypothesis 2: Agreement makes reference to abstract morphosyntactic features.

- ⊙ **Morphosyntactic features** are properties of words that syntax is sensitive to.
- ⊙ H2 thus differs from H1 in that it **centers the abstract feature** (which may determine the morphological form), not the form itself.

■ Relevant feature for agreement in English: number

- ⊙ English has plural and singular number.
- ⇒ the number features of the subject and the verb must match.
- ⊙ Subject [pl] → verb [pl]; subject [sg] → verb [sg]
- ⊙ The **morphological form** of these features is context-dependent!

■ Relevant feature for agreement in English: number

- ⊙ English has plural and singular number.
- ⇒ the number features of the subject and the verb must match.
- ⊙ Subject [pl] → verb [pl]; subject [sg] → verb [sg]
- ⊙ The **morphological form** of these features is context-dependent!

■ It's the features that matter, not the form.

- ⊙ Plural nouns in English come in various morphological forms: *cat* → -s, *ox* → -en, *child* → -ren, *sheep* → ∅, ...
- ⊙ Is there more evidence for H2? Let's look at the verb *take* cross-linguistically!

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin

	SG	PL
1	ná	ná
2	ná	ná
3	ná	ná

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin

	SG	PL
1	ná	ná
2	ná	ná
3	ná	ná

English

	SG	PL
1	take	take
2	take	take
3	takes	take

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

Describe the three paradigms.

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

⇒ Mandarin verbs are **insensitive to the person/number features** of the subject (they do not agree).

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

Describe the three paradigms.

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

⇒ Mandarin verbs are **insensitive to the person/number features** of the subject (they do not agree).

⇒ In German (like English), the verb is **sensitive to the subject's number and person features**.

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

Describe the three paradigms.

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

- ⇒ Mandarin verbs are **insensitive to the person/number features** of the subject (they do not agree).
- ⇒ In German (like English), the verb is **sensitive to the subject's number and person features**.
- ⊙ German has more morphological forms to express the different feature combinations than English, but the system is still pretty simple.

Describe the three paradigms.

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

- ⇒ Mandarin verbs are **insensitive to the person/number features** of the subject (they do not agree).
- ⇒ In German (like English), the verb is **sensitive to the subject's number and person features**.
- ⊙ German has more morphological forms to express the different feature combinations than English, but the system is still pretty simple.

Describe the three paradigms.

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

Describe the three paradigms.

- ⇒ Mandarin verbs are **insensitive to the person/number features** of the subject (they do not agree).
- ⇒ In German (like English), the verb is **sensitive to the subject's number and person features**.
 - ⊙ German has more morphological forms to express the different feature combinations than English, but the system is still pretty simple.

What could a more complex system look like?

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

Describe the three paradigms.

- ⇒ Mandarin verbs are **insensitive to the person/number features** of the subject (they do not agree).
- ⇒ In German (like English), the verb is **sensitive to the subject's number and person features**.
 - ⊙ German has more morphological forms to express the different feature combinations than English, but the system is still pretty simple.

What could a more complex system look like?

- ⊙ Other number feature values cross-linguistically: **dual** (exactly two), **trial** (exactly three) and **paucal** (a few)!

VERBAL AGREEMENT ACROSS LANGUAGES

Mandarin			English		
	SG	PL		SG	PL
1	ná	ná	1	take	take
2	ná	ná	2	take	take
3	ná	ná	3	takes	take

German

	SG	PL
1	nehme	nehmen
2	nimmst	nehmt
3	nimmt	nehmen

Describe the three paradigms.

- ⇒ Mandarin verbs are **insensitive to the person/number features** of the subject (they do not agree).
- ⇒ In German (like English), the verb is **sensitive to the subject's number and person features**.
- ⊙ German has more morphological forms to express the different feature combinations than English, but the system is still pretty simple.

What could a more complex system look like?

- ⊙ Other number feature values cross-linguistically: **dual** (exactly two), **trial** (exactly three) and **paucal** (a few)!
- ⊙ Sensitivity not only to the subject's but the **object's features**, too!

VERBAL AGREEMENT ACROSS LANGUAGES

- Tundra Nenets (Uralic, Russia): agreement with subj's person/num and obj's num:

VERBAL AGREEMENT ACROSS LANGUAGES

■ Tundra Nenets (Uralic, Russia): agreement with subj's person/num and obj's num:

SG object

	SG	DU	PL
1	meə-w ^o	me ^o -m'ih	me ^o -waq
2	meə-r ^o	me ^o -r'ih	me ^o -daq
3	me ^o -da	me ^o -d'ih	me ^o -doh

VERBAL AGREEMENT ACROSS LANGUAGES

■ Tundra Nenets (Uralic, Russia): agreement with subj's person/num and obj's num:

SG object

	SG	DU	PL
1	meə-w ^o	me ^o -m'ih	me ^o -waq
2	meə-r ^o	me ^o -r'ih	me ^o -daq
3	me ^o -da	me ^o -d'ih	me ^o -doh

PL object

	SG	DU	PL
1	me-yə-n ^o	me-y ^o -n'ih	me-y ^o -naq
2	me-yə-d ^o	me-y ^o -d'ih	me-y ^o -daq
3	me-y ^o -da	me-y ^o -d'ih	me-y ^o -doh

VERBAL AGREEMENT ACROSS LANGUAGES

■ Tundra Nenets (Uralic, Russia): agreement with subj's person/num and obj's num:

SG object

	SG	DU	PL
1	meə-w ^o	me ^o -m'ih	me ^o -waq
2	meə-r ^o	me ^o -r'ih	me ^o -daq
3	me ^o -da	me ^o -d'ih	me ^o -doh

PL object

	SG	DU	PL
1	me-yə-n ^o	me-y ^o -n'ih	me-y ^o -naq
2	me-yə-d ^o	me-y ^o -d'ih	me-y ^o -daq
3	me-y ^o -da	me-y ^o -d'ih	me-y ^o -doh

DU object

	SG	DU	PL
1	meŋa-xəyu-n ^o	meŋa-xəyu-n'ih	meŋa-xəyu-naq
2	meŋa-xəyu-d ^o	meŋa-xəyu-d'ih	meŋa-xəyu-daq
3	meŋa-xəyu-da	meŋa-xəyu-d'ih	meŋa-xəyu-doh

AGREEMENT IS ABOUT FEATURES

■ Driving the point home that H2 is the way to go:

AGREEMENT IS ABOUT FEATURES

I Driving the point home that H2 is the way to go:

- ⊙ The complexity of verbal agreement varies immensely across languages.
- ⊙ We want to build a theory of **how the human language faculty works in general**, not merely describe a handful of big languages.

AGREEMENT IS ABOUT FEATURES

I Driving the point home that H2 is the way to go:

- ⊙ The complexity of verbal agreement varies immensely across languages.
- ⊙ We want to build a theory of **how the human language faculty works in general**, not merely describe a handful of big languages.

I Cross-linguistic variability is our greatest advantage!

- ⊙ We want a formal system that can derive the simplicity of Mandarin, the complexity of Tundra Nenets and everything in between.

AGREEMENT IS ABOUT FEATURES

Driving the point home that H2 is the way to go:

- ◉ The complexity of verbal agreement varies immensely across languages.
- ◉ We want to build a theory of **how the human language faculty works in general**, not merely describe a handful of big languages.

Cross-linguistic variability is our greatest advantage!

- ◉ We want a formal system that can derive the simplicity of Mandarin, the complexity of Tundra Nenets and everything in between.
- ◉ Features buy us that system $\Rightarrow$ **Languages vary based on which features of which elements matter** for syntactic processes, such as determining the morphological realization of the verb.

■ So syntax operates on features — but what exactly are they?

■ So syntax operates on features — but what exactly are they?

- ⊙ A feature is a property of a word.
- ⊙ Notation: *word* [feature: value]
- ⊙ Example: *cat* [category: noun, number: singular, animacy: animate, ...]

■ So syntax operates on features — but what exactly are they?

- ⊙ A feature is a property of a word.
- ⊙ Notation: *word* [feature: value]
- ⊙ Example: *cat* [category: noun, number: singular, animacy: animate, ...]

■ Words are feature bundles containing multiple types of features:

- ① phonological features $\Rightarrow$ /kæt/
 - ② semantic features $\Rightarrow$ {animate}
 - ③ morphosyntactic features $\Rightarrow$ {noun, singular}
- $\Rightarrow$ This information is **stored in the mental lexicon**.

- The form of a word adapts to the features in its context.
 - ⊙ This 'adaptation' is called **inflection**.

■ The form of a word adapts to the features in its context.

- ⊙ This 'adaptation' is called **inflection**.
- ⊙ In syntax, there's **nominal inflection** (*declination*) and **verbal inflection** (*conjugation*)
- ⊙ The agreement paradigms we saw are instances of verbal inflection making reference to person and number.

■ The form of a word adapts to the features in its context.

- ⊙ This 'adaptation' is called **inflection**.
- ⊙ In syntax, there's **nominal inflection** (*declination*) and **verbal inflection** (*conjugation*)
- ⊙ The agreement paradigms we saw are instances of verbal inflection making reference to person and number.

■ Morphosyntactic features play a central role in inflection.

■ The form of a word adapts to the features in its context.

- ⊙ This 'adaptation' is called **inflection**.
- ⊙ In syntax, there's **nominal inflection** (*declination*) and **verbal inflection** (*conjugation*)
- ⊙ The agreement paradigms we saw are instances of verbal inflection making reference to person and number.

■ Morphosyntactic features play a central role in inflection.

- ⊙ Are you familiar with other morphosyntactic features?

■ The form of a word adapts to the features in its context.

- ⊙ This 'adaptation' is called **inflection**.
- ⊙ In syntax, there's **nominal inflection** (*declination*) and **verbal inflection** (*conjugation*)
- ⊙ The agreement paradigms we saw are instances of verbal inflection making reference to person and number.

■ Morphosyntactic features play a central role in inflection.

- ⊙ Are you familiar with other morphosyntactic features?
- ⊙ Gender, case, aspect, tense, mood,...

INFLECTIONAL FEATURES

- ⊙ **Number** (singular, plural, dual, trial, paucal)

INFLECTIONAL FEATURES

- ◉ **Number** (singular, plural, dual, trial, paucal)
- ◉ **Person** (first, second, third)
- ◉ Some languages, like Walmajarri (Pama-Nyungan, Australia) have inclusive and exclusive first person (1PL with or without 2SG):

INFLECTIONAL FEATURES

- ◉ **Number** (singular, plural, dual, trial, paucal)
- ◉ **Person** (first, second, third)
- ◉ Some languages, like Walmajarri (Pama-Nyungan, Australia) have inclusive and exclusive first person (1PL with or without 2SG):

	DU	PL
1EXCL	ngayarra	nganimpa
1INCL	ngaliyarra	ngalimpa

INFLECTIONAL FEATURES

- ◉ **Number** (singular, plural, dual, trial, paucal)
- ◉ **Person** (first, second, third)
- ◉ Some languages, like Walmajarri (Pama-Nyungan, Australia) have inclusive and exclusive first person (1PL with or without 2SG):

	DU	PL
1EXCL	ngayarra	nganimpa
1INCL	ngaliyarra	ngalimpa

- ◉ **Gender** (feminine, masculine, neuter, ...) is a subtype of **noun classes**
- ◉ Usually not expressed on the noun itself but on elements agreeing with the noun

INFLECTIONAL FEATURES

- ◉ **Number** (singular, plural, dual, trial, paucal)
- ◉ **Person** (first, second, third)
- ◉ Some languages, like Walmajarri (Pama-Nyungan, Australia) have inclusive and exclusive first person (1PL with or without 2SG):

	DU	PL
1EXCL	ngayarra	nganimpa
1INCL	ngaliyarra	ngalimpa

- ◉ **Gender** (feminine, masculine, neuter, ...) is a subtype of **noun classes**
- ◉ Usually not expressed on the noun itself but on elements agreeing with the noun
- ◉ Fula (Niger-Congo, Western Africa) has 25 noun classes, Hungarian has no noun classes (or gender)!

INFLECTIONAL FEATURES

- ◉ **Number** (singular, plural, dual, trial, paucal)
- ◉ **Person** (first, second, third)
- ◉ Some languages, like Walmajarri (Pama-Nyungan, Australia) have inclusive and exclusive first person (1PL with or without 2SG):

	DU	PL
1EXCL	ngayarra	nganimpa
1INCL	ngaliyarra	ngalimpa

- ◉ **Gender** (feminine, masculine, neuter, ...) is a subtype of **noun classes**
- ◉ Usually not expressed on the noun itself but on elements agreeing with the noun
- ◉ Fula (Niger-Congo, Western Africa) has 25 noun classes, Hungarian has no noun classes (or gender)!

■ **These three features are called ϕ -features (*phi*):**

⇒ They are (among) the central features relevant for inflection in many languages.

A NOTE ON GENDER

■ Grammatical gender is mostly arbitrary!

(5) German

- a. das Mädchen: gender: *neuter*, sexus: *female*
- b. der Tisch gender: *masculine*, sexus: -

■ Grammatical gender is mostly arbitrary!

(5) German

- a. das Mädchen: gender: *neuter*, sexus: *female*
- b. der Tisch gender: *masculine*, sexus: -
- c. das Mädchen, das /*die ich gesehen habe
DET.N girl REL.N REL.F I seen have.1SG
'the girl I have seen'

A NOTE ON GENDER

■ Grammatical gender is mostly arbitrary!

(5) German

- a. das Mädchen: gender: *neuter*, sexus: *female*
- b. der Tisch gender: *masculine*, sexus: -
- c. das Mädchen, das /*die ich gesehen habe
DET.N girl REL.N REL.F I seen have.1SG
'the girl I have seen'

⊙ Syntax is sensitive to the morphosyntactic feature, i.e. grammatical gender.

⊙ **Exception: pronouns!**

- (6) das Mädchen hat sein /ihr Fahrrad geschoben
DET.F girl have.3SG 3SG.POSS.N 3SG.POSS.F bike pushed
'the girl pushed (lit.) its/her bike'

A NOTE ON GENDER

■ Mostly but not entirely arbitrary!

- (7) a. **die** Magd, **der** Tisch, **das** Haus
DET.F maid DET.M table DET.N house
'the maid, the table, the house'

A NOTE ON GENDER

■ Mostly but not entirely arbitrary!

- (7) a. **die** Magd, **der** Tisch, **das** Haus
DET.F maid DET.M table DET.N house
'the maid, the table, the house'
- b. **das** Mäd-chen, **das** Tisch-chen, **das** Häus-chen
DET.N maid-DIM DET.N table-DIM DET.N house-DIM
'the girl (outdated: housemaid), the little table, the little house'

A NOTE ON GENDER

■ Mostly but not entirely arbitrary!

(7) a. **die** Magd, **der** Tisch, **das** Haus
DET.F maid DET.M table DET.N house
'the maid, the table, the house'

b. **das** Mäd-chen, **das** Tisch-chen, **das** Häus-chen
DET.N maid-DIM DET.N table-DIM DET.N house-DIM
'the girl (outdated: housemaid), the little table, the little house'

⊙ In German, the suffixes of derived nouns determine their grammatical gender.

(8) a. eitel, weise, schön
vain wise beautiful

b. **die** Eitel-keit, **die** Weiß-heit, **die** Schön-heit
DET.F vain-KEIT DET.F wise-HEIT DET.F beautiful-HEIT
'the vanity, the wisdom, the beauty'

■ Morphosyntactic features can be divided into two groups:

① Inherent features

- ◇ An element X always bears value Y of feature Z, regardless of syntactic context:
[Z:Y] is a property of X that is stored in the mental lexicon.
- ⇒ Gender of a noun: *Mädchen* is always neuter.

I Morphosyntactic features can be divided into two groups:

① Inherent features

- ◇ An element X always bears value Y of feature Z, regardless of syntactic context: [Z:Y] is a property of X that is stored in the mental lexicon.
- ⇒ Gender of a noun: *Mädchen* is always neuter.

② Assigned features

- ◇ The value of feature Z of element X is determined by the syntactic context (inflection!): the value of Z is not stored in the mental lexicon entry for X.
- ⇒ Gender on an adjective: *klein-es Mädchen, klein-er Tisch*

MORE MORPHOSYNTACTIC FEATURES: CASE

■ Case features are assigned based on syntactic context.

- ◉ Informative about the grammatical function of the element that bears them:

(9) *Hungarian*

Fanni-Ø level-et küld-ött Simon-nak.
Fanni-NOM letter-ACC send-PST.3SG Simon-DAT
'Fanni sent Simon a letter.'

MORE MORPHOSYNTACTIC FEATURES: CASE

| Case features are assigned based on syntactic context.

- ◉ Informative about the grammatical function of the element that bears them:

(9) *Hungarian*

Fanni-Ø level-et küld-ött Simon-nak.
Fanni-NOM letter-ACC send-PST.3SG Simon-DAT
'Fanni sent Simon a letter.'

| Case inventories offer cross-linguistic variability galore:

- ◉ **Mandarin:** no case
- ◉ **German:** four cases (nominative, genitive, accusative, dative)
- ◉ **Hungarian:** around 18 cases

MORE MORPHOSYNTACTIC FEATURES: CASE

Case features are assigned based on syntactic context.

- ◉ Informative about the grammatical function of the element that bears them:

(9) *Hungarian*

Fanni-Ø level-et küld-ött Simon-nak.
Fanni-NOM letter-ACC send-PST.3SG Simon-DAT
'Fanni sent Simon a letter.'

Case inventories offer cross-linguistic variability galore:

- ◉ **Mandarin:** no case
- ◉ **German:** four cases (nominative, genitive, accusative, dative)
- ◉ **Hungarian:** around 18 cases

- ◉ Large inventories encode locative and directional relations via case:

(10) *Hungarian*

a ház-ban, a ház-ba, a ház-ból
DET house-INESS DET house-ILL DET house-ELA
in/into/out of the house

I Tense:

Indicates at what time relative to the time of utterance something happened.

- ⊙ **Past, present, future**
- ⊙ Tense can be expressed **synthetically** (within a single verb: *went*)
- ⊙ Or **periphrastically** (via a combination of multiple verbal elements: *will go*)

MORE MORPHOSYNTACTIC FEATURES: TENSE AND ASPECT

I Tense:

Indicates at what time relative to the time of utterance something happened.

- ⊙ **Past, present, future**
- ⊙ Tense can be expressed **synthetically** (within a single verb: *went*)
- ⊙ Or **periphrastically** (via a combination of multiple verbal elements: *will go*)

I Aspect:

Gives a more **fine-grained ordering** of events within a tense.

- ⊙ **Perfective**: time-constrained, one-time, concluded event
- ⊙ **Imperfective**: continuous, lasting, repeated, non-concluded event

MORE MORPHOSYNTACTIC FEATURES: TENSE AND ASPECT

■ Tense:

Indicates at what time relative to the time of utterance something happened.

- ⊙ **Past, present, future**
- ⊙ Tense can be expressed **synthetically** (within a single verb: *went*)
- ⊙ Or **periphrastically** (via a combination of multiple verbal elements: *will go*)

■ Aspect:

Gives a more **fine-grained ordering** of events within a tense.

- ⊙ **Perfective**: time-constrained, one-time, concluded event
- ⊙ **Imperfective**: continuous, lasting, repeated, non-concluded event

■ These were only a few morphosyntactic features!

- ⊙ Mood, polarity, voice, definiteness, animacy, ...

TYPES OF CONSTITUENTS

■ Recap: What syntax is sensitive to

- ⊙ **Constituents**, (the hierarchical relation between them) and the **features** they carry.
- ⊙ We have seen how the form of a word adapts to the **features in its context** ⇒ **inflection**.

■ Recap: What syntax is sensitive to

- ⊙ **Constituents**, (the hierarchical relation between them) and the **features** they carry.
 - ⊙ We have seen how the form of a word adapts to the **features in its context** ⇒ **inflection**.
 - ⊙ But what determines the context of a word?
- ⇒ **Why are complex structures built in the first place?**

■ Recap: What syntax is sensitive to

- ⊙ **Constituents**, (the hierarchical relation between them) and the **features** they carry.
 - ⊙ We have seen how the form of a word adapts to the **features in its context** ⇒ **inflection**.
 - ⊙ But what determines the context of a word?
- ⇒ **Why are complex structures built in the first place?**
- ⇒ Some constituents have the **requirement to combine** with others.

TYPES OF CONSTITUENTS

■ Recap: What syntax is sensitive to

- ⊙ **Constituents**, (the hierarchical relation between them) and the **features** they carry.
 - ⊙ We have seen how the form of a word adapts to the **features in its context** ⇒ **inflection**.
 - ⊙ But what determines the context of a word?
- ⇒ **Why are complex structures built in the first place?**
- ⇒ Some constituents have the **requirement to combine** with others.

■ Three types of constituents:

- ① **Predicates**: have requirements
- ② **Arguments**: satisfy requirements
- ③ **Adjuncts**: only here for the party

■ Three types of constituents:

I Three types of constituents:

① Predicates:

- ◇ Require additional linguistic material to be complete (they have empty 'places').
- ◇ Express semantic relations between entities.
- ◇ Typically (but not exclusively) verbs $\Rightarrow$ *sleep, eat, hide*

I Three types of constituents:

① Predicates:

- ◇ Require additional linguistic material to be complete (they have empty 'places').
- ◇ Express semantic relations between entities.
- ◇ Typically (but not exclusively) verbs $\Rightarrow$ *sleep, eat, hide*

② Arguments:

- ◇ Complete predicates (occupy their empty 'places').
- ◇ Entities to which the predicate has a relation.
- ◇ Typically (but not exclusively) nouns $\Rightarrow$ *the cat, Hannah, she*

I Three types of constituents:

① Predicates:

- ◇ Require additional linguistic material to be complete (they have empty 'places').
- ◇ Express semantic relations between entities.
- ◇ Typically (but not exclusively) verbs ⇒ *sleep, eat, hide*

② Arguments:

- ◇ Complete predicates (occupy their empty 'places').
- ◇ Entities to which the predicate has a relation.
- ◇ Typically (but not exclusively) nouns ⇒ *the cat, Hannah, she*

③ Adjuncts:

- ◇ Optional – the sentence is grammatical with or without them.
- ◇ Think of them as embellishments ⇒ *in the cabinet, at 10 in the morning*

- Predicates trigger processes to build syntactic structure by requiring arguments.

¹More on zero-place predicates in session 4!

■ Predicates trigger processes to build syntactic structure by requiring arguments.

- ① one-place (**intransitive**) predicates:
fall, work, arrive, ... $\Rightarrow$ **The train** arrived.

¹More on zero-place predicates in session 4!

■ Predicates trigger processes to build syntactic structure by requiring arguments.

- ① one-place (**intransitive**) predicates:
fall, work, arrive, ... ⇒ **The train** arrived.
- ② two-place (**transitive**) predicates:
believe, love, devour, ... ⇒ **The mouse** devoured **the cheese**.

¹More on zero-place predicates in session 4!

■ Predicates trigger processes to build syntactic structure by requiring arguments.

- ① one-place (**intransitive**) predicates:
fall, work, arrive, ... ⇒ **The train** arrived.
- ② two-place (**transitive**) predicates:
believe, love, devour, ... ⇒ **The mouse** devoured **the cheese**.
- ③ three-place (**ditransitive**):
give, show, offer, ... ⇒ **Aladdin** showed **Jasmine** **the world**.

¹More on zero-place predicates in session 4!

■ Predicates trigger processes to build syntactic structure by requiring arguments.

- ① one-place (**intransitive**) predicates:
fall, work, arrive, ... ⇒ **The train** arrived.
- ② two-place (**transitive**) predicates:
believe, love, devour, ... ⇒ **The mouse** devoured **the cheese**.
- ③ three-place (**ditransitive**):
give, show, offer, ... ⇒ **Aladdin** showed **Jasmine** **the world**.
- ④ zero-place predicates:
rain, snow, ... ⇒ **It** rained.¹

¹More on zero-place predicates in session 4!

■ Predicates trigger processes to build syntactic structure by requiring arguments.

- ① one-place (**intransitive**) predicates:
fall, work, arrive, ... ⇒ **The train** arrived.
- ② two-place (**transitive**) predicates:
believe, love, devour, ... ⇒ **The mouse** devoured **the cheese**.
- ③ three-place (**ditransitive**):
give, show, offer, ... ⇒ **Aladdin** showed **Jasmine** **the world**.
- ④ zero-place predicates:
rain, snow, ... ⇒ **It** rained.¹

■ **Valency: the number of arguments a predicate needs to express a meaningful event.**

¹More on zero-place predicates in session 4!

- Arguments play a specific role in the event described by the predicate.

- Arguments play a specific role in the event described by the predicate.
 - ⊙ This role is called a θ -role (*theta*) $\Rightarrow$ predicates **assign** θ -roles to their arguments.

■ Arguments play a specific role in the event described by the predicate.

- ⊙ This role is called a θ -role (*theta*) $\Rightarrow$ predicates **assign** θ -roles to their arguments.

■ Some important θ -roles

- ① **Agent** or **causer** or **actor** $\Rightarrow$ the active performer of the action
He is working./The mouse devoured the cheese.

Arguments play a specific role in the event described by the predicate.

- ⊙ This role is called a θ -role (*theta*) $\Rightarrow$ predicates **assign** θ -roles to their arguments.

Some important θ -roles

- ① **Agent** or **causer** or **actor** $\Rightarrow$ the active performer of the action
He is working./The mouse devoured the cheese.
- ② **Theme** or **patient** $\Rightarrow$ the entity affected by the action
He collapsed./The mouse devoured the cheese./I gave her books.

Arguments play a specific role in the event described by the predicate.

- ⊙ This role is called a θ -role (*theta*) $\Rightarrow$ predicates **assign** θ -roles to their arguments.

Some important θ -roles

- ① **Agent** or **causer** or **actor** $\Rightarrow$ the active performer of the action
He is working./The mouse devoured the cheese.
- ② **Theme** or **patient** $\Rightarrow$ the entity affected by the action
*He collapsed./The mouse devoured **the cheese**./I gave her **books**.*
- ③ **Goal** $\Rightarrow$ the goal/endpoint of the action
*I gave **her** books.*

Arguments play a specific role in the event described by the predicate.

- ◉ This role is called a θ -role (*theta*) $\Rightarrow$ predicates **assign** θ -roles to their arguments.

Some important θ -roles

- 1 **Agent** or **causer** or **actor** $\Rightarrow$ the active performer of the action
He is working./The mouse devoured the cheese.
- 2 **Theme** or **patient** $\Rightarrow$ the entity affected by the action
*He collapsed./The mouse devoured **the cheese**./I gave her **books**.*
- 3 **Goal** $\Rightarrow$ the goal/endpoint of the action
*I gave **her** books.*
- 4 **Experiencer** $\Rightarrow$ the entity experiencing/feeling something
***Trixie** felt hot./**He** was scared.*

CONDITIONS ON θ -ROLE ASSIGNMENT

I θ -role assignment

- ⊙ θ -roles go hand in hand with a predicate's valency.

⇒ **For each argument a predicate requires, it has a θ -role to assign.**

CONDITIONS ON θ -ROLE ASSIGNMENT

■ θ -role assignment

- ◉ θ -roles go hand in hand with a predicate's valency.

⇒ **For each argument a predicate requires, it has a θ -role to assign.**

■ Theta-criterion:

- ◉ Each θ -role must be assigned to an argument (discharging it does not suffice):

(11) *Arrives.

(12) *Eats the cheese.

CONDITIONS ON θ -ROLE ASSIGNMENT

■ θ -role assignment

- ◉ θ -roles go hand in hand with a predicate's valency.

⇒ **For each argument a predicate requires, it has a θ -role to assign.**

■ Theta-criterion:

- ◉ Each θ -role must be assigned to an argument (discharging it does not suffice):

(11) *Arrives.

(12) *Eats the cheese.

- ◉ Each argument is assigned one and only one θ -role.

(13) *Freddie likes. ⇒ Cannot mean *Freddie likes himself*.

CONDITIONS ON θ -ROLE ASSIGNMENT

■ θ -role assignment

- ◉ θ -roles go hand in hand with a predicate's valency.

⇒ **For each argument a predicate requires, it has a θ -role to assign.**

■ Theta-criterion:

- ◉ Each θ -role must be assigned to an argument (discharging it does not suffice):

(11) *Arrives.

(12) *Eats the cheese.

- ◉ Each argument is assigned one and only one θ -role.

(13) *Freddie likes. ⇒ Cannot mean *Freddie likes himself*.

■ **There are some apparent counterexamples — you can find them in the appendix.**

I Refining morphosyntactic features

- ⊙ The morphosyntactic features we saw so far determined how the elements in their context inflect $\Rightarrow$ they are called **inflectional features**.

I Refining morphosyntactic features

- ⊙ The morphosyntactic features we saw so far determined how the elements in their context inflect $\Rightarrow$ they are called **inflectional features**.
- ⊙ Another type of morphosyntactic feature is the **category feature**: it is an inherent feature stored in the mental lexicon encoding what word category a lexical element belongs to.

DIFFERENT FEATURES DO DIFFERENT THINGS

I Refining morphosyntactic features

- ⊙ The morphosyntactic features we saw so far determined how the elements in their context inflect $\Rightarrow$ they are called **inflectional features**.
- ⊙ Another type of morphosyntactic feature is the **category feature**: it is an inherent feature stored in the mental lexicon encoding what word category a lexical element belongs to.

I Why do we need category features?

DIFFERENT FEATURES DO DIFFERENT THINGS

I Refining morphosyntactic features

- ⊙ The morphosyntactic features we saw so far determined how the elements in their context inflect $\Rightarrow$ they are called **inflectional features**.
- ⊙ Another type of morphosyntactic feature is the **category feature**: it is an inherent feature stored in the mental lexicon encoding what word category a lexical element belongs to.

I Why do we need category features?

- ⊙ Predicates do not just combine with any argument, they require their arguments to be of a specific category.
- ⊙ This choosiness is called **c-selection** or **subcategorization**.

■ Some predicates require nominal objects:

- (14) a. Barry interrogated [the suspect].
b. *Barry interrogated [at the suspect].
c. *Barry interrogated [screaming].
d. *Barry interrogated [whether she knows the suspect].

■ Some predicates require nominal objects:

- (14) a. Barry interrogated [the suspect].
b. *Barry interrogated [at the suspect].
c. *Barry interrogated [screaming].
d. *Barry interrogated [whether she knows the suspect].

■ Some others require sentential objects:

- (15) a. Carla admitted [that Helen was lovely].
b. *Carla admitted [linguistics].

■ Yet others require prepositional objects:

- (16) a. Harry is obsessed [with the Beatles].
b. *Harry is obsessed [the Beatles].
c. *Harry is obsessed [that the Beatles are great].

■ Yet others require prepositional objects:

- (16) a. Harry is obsessed [with the Beatles].
b. *Harry is obsessed [the Beatles].
c. *Harry is obsessed [that the Beatles are great].

■ And a few predicates are happy with an argument with either category feature:

- (17) a. Ringo believes [him].
b. Ringo believes [in the power of love].
c. Ringo believes [that Celine's heart will go on].

C-SELECTION FEATURES

I C-selection feature notation

- ⊙ We will note c-selection features like this: **[uX]** (**X = variable over category**)
- ⊙ The **number of [uX]-features corresponds to the valency** of the predicate (how many arguments it takes/how many 'places' it has)

C-SELECTION FEATURES

■ C-selection feature notation

- ⊙ We will note c-selection features like this: $[uX]$ (X = variable over category)
- ⊙ The **number of $[uX]$ -features corresponds to the valency** of the predicate (how many arguments it takes/how many 'places' it has)

■ Each predicate has a category feature and a c-selection feature.

- (18) a. *laugh* $[V, uN]$
→ intransitive predicate requiring **one nominal argument**
- b. *see* $[V, uN, uN]$
→ transitive predicate requiring **two nominal arguments**

ARGUMENT STRUCTURE

■ The argument structure specification of predicates could be depicted like this:

- ⊙ It contains the predicate, the θ -roles it assigns and its c-selection features.

Predicate	θ -grid	c-selection	Example
work	agent	[uN]	Claudia works.
love			

ARGUMENT STRUCTURE

■ The argument structure specification of predicates could be depicted like this:

- ⊙ It contains the predicate, the θ -roles it assigns and its c-selection features.

Predicate	θ -grid	c-selection	Example
work	agent	[uN]	Claudia works.
love	experiencer	[uN]	Claudia loves Tina.
	patient	[uN]	
send			

ARGUMENT STRUCTURE

■ The argument structure specification of predicates could be depicted like this:

- ⊙ It contains the predicate, the θ -roles it assigns and its c-selection features.

Predicate	θ -grid	c-selection	Example
work	agent	[uN]	Claudia works.
love	experiencer	[uN]	Claudia loves Tina.
	patient	[uN]	
send	agent	[uN]	Claudia sent a letter to Tina.
	patient	[uN]	
	goal	[uP]	

■ The argument structure of a predicate determines:

- ① how many arguments ($\Rightarrow$ valency)
- ② of which category ($\Rightarrow$ c-selection)
- ③ (in which order) **a predicate has to combine with to be *saturated*.**

It also determines:

- ④ the semantic relation the predicate has with its argument(s) ($\Rightarrow$ θ -role assignment).

■ The argument structure of a predicate determines:

- ① how many arguments ($\Rightarrow$ valency)
- ② of which category ($\Rightarrow$ c-selection)
- ③ (in which order) **a predicate has to combine with to be saturated.**

It also determines:

- ④ the semantic relation the predicate has with its argument(s) ($\Rightarrow$ θ -role assignment).
- The argument structure of a predicate is **stored in the mental lexicon.**
 - That means it is not computed on the spot and therefore **unpredictable.**

THIS WAS A LOT, BUT IT WILL CARRY US REALLY FAR.

■ It wouldn't surprise me if you thought syntax was a pain in the butt by now.

⊙ **Syntax has a lot of requirements to satisfy!**

- ◇ Lexical elements need to inflect for the abstract features in their context,
- ◇ languages differ wildly regarding in which and to which extent these features matter,
- ◇ predicates need certain arguments and need to assign all of their θ -roles, ...

THIS WAS A LOT, BUT IT WILL CARRY US REALLY FAR.

■ It wouldn't surprise me if you thought syntax was a pain in the butt by now.

- ⊙ **Syntax has a lot of requirements to satisfy!**

- ◇ Lexical elements need to inflect for the abstract features in their context,
- ◇ languages differ wildly regarding in which and to which extent these features matter,
- ◇ predicates need certain arguments and need to assign all of their θ -roles, ...

- ⊙ This is precisely the point: our language faculty's computational system is **incredibly powerful and creative**, but this power and creativity must also be constrained enough to allow us to **communicate effectively** with one another.

■ Next up: satisfying all of these requirements via structure building!

Ask away!

- (19) a. Dolly donated half a million dollars.
b. Charlie is presenting this week.

■ Making sense of (19)

- ⊙ To perceive these sentences as grammatical, it must be contextually clear what the goal of *donate*/the theme of *present* is.
- ⊙ It could be that the element in question is present in the syntactic structure and simply not realized phonologically.
- ⊙ Not all predicates allow for this, and we will set such cases aside for now.

■ Predicates can also select the semantic properties of their arguments.

- ⊙ For example, some predicates require (in)animate arguments:

- (20) a. #²The wall is running.
b. #The apple is laughing.
c. #Colourless green ideas sleep furiously.

⇒ This is called s-selection (s for semantic).

- ⊙ Syntactically, there is nothing wrong with these examples, but they are semantically ill-formed.

²We mark semantically deviant examples with #.